

KChat On-Device Model Strategy Proposal v3

Target platforms

- A. Android
- B. iOS, with MLX optimization where possible
- C. MacBook/macOS, with MLX optimization where possible
- D. Windows laptop, explicitly split into CPU-only and CPU+GPU paths

Capability scope

- Understanding: guardrail, classification, semantic search, OCR, speech-to-text, image/video understanding.
- Generation: text generation, summarization, translation, drafting, task extraction, image generation, music/vocal generation, video generation.
- Principle: minimal core by default; large creative models only as optional add-ons.

1. Audit of previous proposal

1.1 Main issues to correct

Issue	Correction
The previous proposal was too cross-platform and did not follow the requested platform-by-platform structure.	This rewrite is organized as Android, iOS, MacBook, and Windows, each with low/mid/high tier decisions.
iOS and MacBook were partially conflated.	iOS uses MLX Swift only where stable and memory-safe; MacBook Apple Silicon can use MLX more broadly.
Windows was not explicit enough about CPU-only versus CPU+GPU.	Windows is now split per tier into CPU-only and CPU+GPU profiles.
Creative add-ons were too broadly attached to mobile.	Image generation is possible on high-end mobile; music/vocal and video generation are desktop-first.

Issue	Correction
Some model names were too generic.	This rewrite names actual models and distinguishes core, fallback, candidate replacement, and add-on models.
Low-end generation was not treated carefully.	Low-end devices get a small text generator only if benchmarks pass; creative generation is disabled.
“MLX optimization” needed precision.	MLX/MLX Swift is used for Apple Silicon text/VLM workloads where possible; Core ML remains preferred for ANE-oriented classifier, embedding, and diffusion deployment.

The existing KChat strategy is still a good base. It already defines XLM-R for guardrail/search, MobileNetV3-Small and TinyViT-5M for media safety, a Windows-only OCR pack, Bonsai-1.7B for chat/policy interpretation, MobileCLIP-S2 for media search, and Whisper Tiny/Base for transcription. It also already documents INT8/INT4 sizes, lazy loading, cross-repo sharing, and CPU-only Windows constraints. ([GitHub](#))

1.2 Core strategy decision

The new strategy should keep the existing **minimal shared core**:

- One multilingual text encoder: **XLM-R**.
- One default small language model: **Bonsai-1.7B**.
- One fallback small language model: **Qwen3-0.6B**.
- One base image safety classifier: **MobileNetV3-Small**.
- One enhanced image safety classifier: **TinyViT-5M**.
- One media embedding model: **MobileCLIP2-S2** if validated, otherwise **MobileCLIP-S2**.
- One speech-to-text ladder: **Whisper Tiny** and **Whisper Base**.
- One optional VLM: **SmolVLM2-500M-Video-Instruct**.
- Creative add-ons are separate packs.

The current KChat document already deduplicates XLM-R between text guardrail and semantic search and shares Bonsai-1.7B between chat and CV-Guard policy interpretation. That deduplication should remain mandatory. ([GitHub](#))

2. Proposed model pack structure

2.1 Core packs

Pack	Required?	Purpose	Actual models
Core A: Safety bootstrap	Yes	Immediate safety before downloads	MobileNetV3-Small INT8, KChat skill pack, rule fallback
Core B: Text understanding	Yes	guardrail, search, classification	KChat XLM-R derived from FacebookAI/xlm-roberta-base, xlmr.onnx, xlmr.int4.onnx, xlmr_head.npz, xlmr.spm
Core C: Text generation	Yes, tier-gated	chat, summary, translation, drafting, extraction	prism-ml/Ternary-Bonsai-1.7B-gguf, prism-ml/Ternary-Bonsai-1.7B-mlx-2bit, fallback Qwen/Qwen3-0.6B
Core D: Media safety	Yes	image/video safety	MobileNetV3-Small, timm/tiny_vit_5m_224.dist_in22k_ft_in1k
On-demand E: Speech	No	speech-to-text and optional TTS	openai/whisper-tiny, openai/whisper-base, optional onnx-community/Kokoro-82M-v1.0-ONNX
On-demand F: Media understanding	No	image/video search, VQA, captions	apple/MobileCLIP-S2 or apple/MobileCLIP2-S2, optional HuggingFaceTB/SmolVLM2-500M-Video-Instruct

Bonsai-1.7B is an appropriate default SLM because the GGUF version is a Qwen3-derived 1.72B-parameter model with a packed Q2_0 size around 436 MiB, while the MLX 2-bit version is built for Apple Silicon and reports a packed size around 0.45 GiB. ([Hugging Face](#))

Qwen3-0.6B is the fallback model because it has 0.6B parameters and a 32,768-token context, making it a better fit for low-end or CPU-only devices than a 1.7B model. ([Hugging Face](#))

2.2 Creative add-on packs

Add-on	Required?	Purpose	Actual models
Add-on G: Image generation	No	text-to-image, image-to-image	stabilityai/sd-turbo, stabilityai/sd-xl-turbo; Apple deployment via Core ML Stable Diffusion
Add-on H: Music + vocal generation	No	sound loops, songs, vocals	stabilityai/stable-audio-open-small, ACE-Step/ACE-Step-v1-3.5B, optional HKUSTAudio/YuE-s2-1B-general, m-a-p/YuE-s1-7B-anneal-en-cot
Add-on I: Video generation	No	text-to-video, image-to-video	Wan-AI/Wan2.1-T2V-1.3B, llyasviel/FramePack, hunyuanvideo-community/HunyuanVideo, Lightricks/LTX-Video, Wan-AI/Wan2.2-TI2V-5B, Wan-AI/Wan2.2-T2V-A14B

SD-Turbo and SDXL-Turbo are appropriate image add-on candidates because both are distilled text-to-image models designed for one-to-four-step generation; SD-Turbo is lighter, while SDXL-Turbo gives better quality and prompt following. ([Hugging Face](#))

Music and video add-ons are not core. Stable Audio Open Small generates up to 11 seconds of stereo audio at 44.1 kHz, ACE-Step targets full-song music generation, and YuE targets lyrics-to-song full-song generation with vocals and accompaniment. ([Hugging Face](#))

Video generation must be desktop-first. Wan2.1-T2V-1.3B reports an 8.19 GB VRAM requirement for consumer GPUs; FramePack requires an NVIDIA RTX 30/40/50-series GPU and at least 6 GB GPU memory; LTX Desktop requires ≥ 16 GB VRAM for local generation; Wan2.2-T2V-A14B documents an 80 GB VRAM command for 720p generation. ([Hugging Face](#))

3. Global model policy

3.1 Core model selection

Function	Low tier	Mid tier	High tier
Text understanding	XLM-R INT4	XLM-R INT4 or INT8	XLM-R INT8
Text generation	Qwen3-0.6B or Bonsai-1.7B only if benchmark passes	Bonsai-1.7B	Bonsai-1.7B; optional Qwen3-4B add-on
Media safety	MobileNetV3-Small only	MobileNetV3-Small + TinyViT-5M	MobileNetV3-Small + TinyViT-5M
OCR	Platform OCR where available	Platform OCR	Platform OCR + optional OCR pack
Speech-to-text	Whisper Tiny	Whisper Tiny/Base	Whisper Base
Media search	MobileCLIP-S2 INT4	MobileCLIP-S2 or MobileCLIP2-S2 INT8	MobileCLIP2-S2 INT8
Visual Q&A	disabled	optional SmolVLM2	SmolVLM2 or Gemma 3n candidate
Image generation	disabled	limited/experimental	optional add-on
Music/vocal generation	disabled	desktop only	desktop only
Video generation	disabled	desktop GPU only	desktop GPU/workstation only

MobileCLIP2-S2 is a candidate upgrade over MobileCLIP-S2 because Apple’s model card shows better zero-shot ImageNet performance than MobileCLIP-S2 at the same S2 checkpoint class, but KChat should not replace S2 until regression tests confirm retrieval quality and export/runtime stability. ([Hugging Face](#))

3.2 Quantization rules

Model family	Low tier	Mid tier	High tier
XLM-R	INT4 ONNX	INT4 or INT8 ONNX	INT8 ONNX

Model family	Low tier	Mid tier	High tier
Bonsai-1.7B	Q2_0 GGUF / 2-bit MLX if benchmark passes	Q2_0 GGUF / 2-bit MLX	Q2_0 GGUF / 2-bit MLX
Qwen3-0.6B	Q4_K_M GGUF	fallback only	not default
Qwen3-4B-Instruct-2507	disabled	disabled by default	optional Q4/Q5 quality pack
MobileNetV3-Small	INT8	INT8	INT8
TinyViT-5M	disabled or INT8	INT8	INT8
MobileCLIP	INT4 storage-constrained	INT8	INT8
Whisper	INT8	INT8	INT8 / FP16 if faster
SD-Turbo / SDXL-Turbo	disabled	FP16 / Core ML compressed only after validation	FP16, Core ML palettized/quantized where validated
Audio/video diffusion	disabled	FP16/BF16 GPU only	FP16/BF16, selective 8-bit/4-bit only after quality tests

ONNX Runtime should remain the common runtime for encoder/classifier models because it supports many execution providers, including CPU, XNNPACK, Core ML, DirectML, CUDA, and others. ([ONNX Runtime](#))

4. A — Android proposal

Android should use **LiteRT/TFLite for new vision and diffusion work, ONNX Runtime Mobile or LiteRT-converted models for encoders**, and **llama.cpp/GGUF for SLM generation**. NNAPI should not be the primary strategy because Android’s NNAPI is deprecated in Android 15; LiteRT is Google’s current on-device ML/GenAI framework. ([Android Developers](#))

A1. Android low tier

Target device

Parameter	Target
RAM	2–4 GB
Storage headroom	0.5–1.5 GB
Compute	CPU-first; weak GPU/NPU
Examples	budget Android, older Snapdragon/Dimensity
Default mode	understanding-first, short generation only

Core models

Capability	Model	Format	Quantization	Runtime
Text guardrail/search	KChat XLM-R	ONNX	INT4	ONNX Runtime Mobile CPU/XNNPACK
Text generation	Qwen/Qwen3-0.6B	GGUF	Q4_K_M	llama.cpp NDK CPU
Text generation alternative	prism-ml/Ternary-Bonsai-1.7B-gguf	GGUF	Q2_0	only if benchmark passes
Image safety	MobileNetV3-Small	TFLite/LiteRT	INT8	LiteRT CPU/GPU
Enhanced safety	none	n/a	n/a	disabled
OCR	ML Kit Text Recognition	system	n/a	Android API
Speech-to-text	openai/whisper-tiny	ONNX / whisper.cpp	INT8	CPU
Media search	apple/MobileCLIP-S2	ONNX	INT4	lazy download
Visual Q&A	none	n/a	n/a	disabled
TTS	Android system TTS	system	n/a	system

Generation capability

Task	Enabled? Limits	
Smart reply	Yes	32–96 tokens
Short summary	Yes	128 tokens
Translation	Yes	short message only
Drafting	Limited	short drafts
Task extraction	Yes	grammar-constrained JSON
Image generation	No	disabled
Music/vocal generation	No	disabled
Video generation	No	disabled

Optimization

- Use short context windows: 1K–2K tokens.
- Use retrieval instead of long-context prompting.
- Disable thinking mode for Qwen3 unless the user explicitly asks for reasoning-heavy output.
- Keep one SLM loaded at a time.
- Unload the SLM after 30–60 seconds idle.
- Do not download MobileCLIP or Whisper until first use.

Add-on policy

Add-on	Decision
Image generation	Not offered
Music/vocal generation	Not offered
Video generation	Not offered
VLM understanding	Not offered

A2. Android mid tier

Target device

Parameter	Target
RAM	6–8 GB
Storage headroom	1.5–4 GB
Compute	CPU + GPU/NPU where stable
Examples	Snapdragon 7-class, Dimensity 8000-class
Default mode	full text understanding + normal text generation

Core models

Capability	Model	Format	Quantization	Runtime
Text guardrail/sear ch	KChat XLM-R	ONNX	INT4 or INT8	ONNX Runtime Mobile / LiteRT conversion if validated
Text generation	prism-ml/Ternary-Bonsai-1.7B-gguf	GGUF	Q2_0	llama.cpp NDK
Fallback SLM	Qwen/Qwen3-0.6B	GGUF	Q4_K_M	llama.cpp NDK
Image safety	MobileNetV3-Small	LiteRT/TFLite	INT8	LiteRT
Enhanced safety	timm/tiny_vit_5m_224.dist_in22k_ft_in1k	LiteRT/TFLite/ONNX	INT8	LiteRT preferred
OCR	ML Kit Text Recognition	system	n/a	Android API

Capability	Model	Format	Quantization	Runtime
Speech-to-text	openai/whisper-tiny default; openai/whisper-base if benchmark passes	ONNX / whisper.cpp	INT8	CPU/GPU where stable
Media search	apple/MobileCLIP-S2 or validated apple/MobileCLIP2-S2	ONNX	INT8	lazy download
Visual Q&A	HuggingFaceTB/SmoVLM2-500M- Video-Instruct	ONNX/GGUF candidate	4-bit/8-bit	optional, benchmark- gated
TTS	Android system TTS; optional Kokoro later	system / ONNX	q8/q4	optional

SmoVLM2-500M is suitable as an optional media-understanding pack because it processes videos, images, and text inputs to produce text outputs, and its model card positions it for on-device applications with limited compute. It does not generate images or videos. ([Hugging Face](#))

Generation capability

Task	Enabled?	Limits
Smart reply	Yes	64–256 tokens
Summary	Yes	256–512 tokens
Translation	Yes	message/thread fragments
Drafting	Yes	512 tokens default
Task extraction	Yes	grammar-constrained JSON
Image generation	Experimental only stabilityai/sd-turbo, 512×512, user opt-in	
Music/vocal generation	No	disabled
Video generation	No	disabled

Add-on policy

Add-on	Decision
Image generation	Optional experiment only if ≥ 8 GB RAM, ≥ 4 GB storage, thermal-normal
Music/vocal generation	Not offered
Video generation	Not offered
VLM understanding	Optional

A3. Android high-end tier

Target device

Parameter	Target
RAM	12–16 GB
Storage headroom	4–12 GB
Compute	flagship GPU/NPU
Examples	Snapdragon 8-class, Dimensity 9000-class
Default mode	full text + media understanding; optional image generation

Core models

Capability	Model	Format	Quantization	Runtime
Text guardrail/search	KChat XLM-R	ONNX	INT8	ONNX Runtime Mobile / LiteRT if validated
Text generation	prism-ml/Ternary-Bonsai-1.7B-gguf	GGUF	Q2_0	llama.cpp NDK

Capability	Model	Format	Quantization	Runtime
Optional quality SLM	Qwen/Qwen3-4B-Instruct-2507	GGUF	Q4_K_M / Q5_K_M	opt-in only
Image safety	MobileNetV3-Small + TinyViT-5M	LiteRT/TFLite/ONNX	INT8	LiteRT preferred
OCR	ML Kit Text Recognition	system	n/a	Android API
Speech-to-text	openai/whisper-base	ONNX / whisper.cpp	INT8	CPU/GPU
Media search	apple/MobileCLIP2-S2 preferred after validation	ONNX	INT8	lazy
Visual Q&A	HuggingFaceTB/SmolVLM2-500M-Video-Instruct; candidate google/gemma-3n-E2B-it	ONNX/GGUF/Transformers-derived	4-bit/8-bit	optional
TTS	Android system TTS or onnx-community/Kokoro-82M-v1.0-ONNX	ONNX	q8/q4	optional

Gemma 3n E2B is a candidate replacement for SmolVLM2 only on high-end devices because it supports text, audio, image, and video inputs while generating text outputs, but it has higher complexity and should not be stacked redundantly with SmolVLM2. ([Hugging Face](#))

Generation add-ons

Add-on	Model	Decision
Image generation	stabilityai/sd-turbo	allowed after benchmark

Add-on	Model	Decision
Higher image generation	stabilityai/sd-xl-turbo	generally not default on Android; experimental only
Music/vocal generation	none	disabled
Video generation	none	disabled

Optimization

- Keep text generation and image generation mutually exclusive in memory.
- Use 512×512 image output for SD-Turbo.
- Use one-to-four-step inference.
- Run generated-image safety scan after generation.
- Disable background image generation on low battery or high thermal state.

5. B — iOS proposal, with MLX optimization where possible

iOS should use **Core ML** for classifiers, encoders, image models, and ANE-oriented workloads. Use **MLX Swift** for LLM/VLM generation only where device support, memory behavior, and App Store packaging are proven. Apple’s MLX Swift examples include reusable LLM, VLM, embedding, and Stable Diffusion libraries, but Core ML remains the more mature path for ANE-targeted deployment. ([GitHub](#))

B1. iOS low tier

Target device

Parameter	Target
RAM	3–4 GB effective app memory
Storage headroom	0.5–1.5 GB
Compute	older A-series CPU/GPU/Neural Engine
Default mode	understanding-first, short text generation only

Core models

Capability	Model	Format	Quantization	Runtime
Text guardrail/search	KChat XLM-R	Core ML or ONNX	INT4	Core ML / ONNX Runtime CoreML EP
Text generation	Qwen/Qwen3-0.6B	MLX or GGUF	4-bit	MLX Swift if stable; llama.cpp Metal/CPU fallback
Text generation alternative	prism-ml/Ternary-Bonsai-1.7B-mlx-2bit	MLX	2-bit	only if benchmark passes
Image safety	MobileNetV3-Small	Core ML	INT8 / palettized	Core ML
Enhanced safety	none	n/a	n/a	disabled
OCR	Apple Vision	system	n/a	Vision framework
Speech-to-text	openai/whisper-tiny	Core ML / MLX / whisper.cpp	INT8	benchmark-selected
Media search	apple/MobileCLIP-S2	Core ML	INT4	lazy
Visual Q&A	none	n/a	n/a	disabled
TTS	iOS system TTS	system	n/a	system

Generation capability

Task	Enabled?	Limits
Smart reply	Yes	32–96 tokens
Short summary	Yes	128 tokens
Translation	Limited	short messages
Drafting	Limited	short drafts
Image generation	No	disabled

Task	Enabled? Limits	
Music/vocal generation	No	disabled
Video generation	No	disabled

MLX policy

- Use MLX Swift only after a device benchmark.
- Keep Core ML for safety classifiers and embeddings.
- Do not force Bonsai on low-memory iPhones.
- Use iOS memory-pressure callbacks to unload SLM immediately.

B2. iOS mid tier

Target device

Parameter	Target
RAM	4–8 GB effective
Storage headroom	1.5–4 GB
Compute	recent A-series CPU/GPU/Neural Engine
Default mode	text understanding + text generation + speech

Core models

Capability	Model	Format	Quantization	Runtime
Text guardrail/search	KChat XLM-R	Core ML / ONNX	INT4 or INT8	Core ML / ONNX Runtime
Text generation	prism-ml/Ternary-Bonsai-1.7B-mlx-2bit	MLX	2-bit	MLX Swift
Fallback SLM	Qwen/Qwen3-0.6B	MLX/GGUF	4-bit	fallback

Capability	Model	Format	Quantization	Runtime
Image safety	MobileNetV3-Small	Core ML	INT8	Core ML
Enhanced safety	timm/tiny_vit_5m_224.dist_in22k_ft_in1k	Core ML	INT8 / palettized	Core ML
OCR	Apple Vision	system	n/a	Vision
Speech-to-text	openai/whisper-tiny or openai/whisper-base	Core ML / MLX / whisper.cpp	INT8	benchmark-selected
Media search	apple/MobileCLIP-S2 or apple/MobileCLIP2-S2	Core ML	INT8	lazy
Visual Q&A	HuggingFaceTB/SmolVLM2-500M-Video-Instruct	MLX/VLM candidate	4-bit/8-bit	optional
TTS	system TTS or Kokoro ONNX if runtime available	system/ONNX	q8/q4	optional

Generation add-ons

Add-on	Model	Decision
Image generation	stabilityai/sd-turbo via Core ML conversion	optional only
Higher image generation	stabilityai/sd-xl-turbo	not default
Music/vocal generation	none	disabled
Video generation	none	disabled

For image models on iPhone/iPad, use Apple’s Core ML Stable Diffusion conversion path and prefer SPLIT_EINSUM for cpuAndNeuralEngine deployment; Apple’s own guidance also recommends smaller latent sizes for iOS/iPadOS deployment rather than default full SDXL resolution. ([GitHub](#))

B3. iOS high-end tier

Target device

Parameter	Target
RAM	8 GB+ effective
Storage headroom	4–12 GB
Compute	newer Pro iPhone/iPad CPU/GPU/Neural Engine
Default mode	full local text + speech + visual understanding; optional image generation

Core models

Capability	Model	Format	Quantization	Runtime
Text guardrail/search	KChat XLM-R	Core ML / ONNX	INT8	Core ML / ONNX Runtime
Text generation	prism-ml/Ternary-Bonsai-1.7B-mlx-2bit	MLX	2-bit	MLX Swift
Optional quality SLM	Qwen/Qwen3-4B-Instruct-2507	MLX	4-bit	opt-in, benchmark-gated
Image safety	MobileNetV3-Small + TinyViT-5M	Core ML	INT8 / palettized	Core ML
OCR	Apple Vision	system	n/a	Vision
Speech-to-text	openai/whisper-base	Core ML / MLX / whisper.cpp	INT8	benchmark-selected
Media search	apple/MobileCLIP2-S2	Core ML	INT8	lazy
Visual Q&A	SmolVLM2-500M-Video-Instruct; candidate Gemma 3n E2B	MLX/VLM candidate	4-bit/8-bit	optional

Capability	Model	Format	Quantization	Runtime
TTS	onnx-community/Kokoro-82M-v1.0-ONNX or system TTS	ONNX/system	q8/q4	optional

Qwen3-4B-Instruct-2507 is only a high-end optional model because it has 4.0B parameters and a native 262,144-token context; in mobile deployment, that context must be capped aggressively to avoid memory pressure. ([Hugging Face](#))

Generation add-ons

Add-on	Model	Decision
Image generation	stabilityai/sd-turbo	allowed
Higher image generation	stabilityai/sdxl-turbo via Core ML experimental	
Music/vocal generation	none	disabled by default
Video generation	none	disabled by default

MLX optimization

- Use prism-ml/Ternary-Bonsai-1.7B-mlx-2bit as the preferred SLM.
- Use MLX Swift streaming generation.
- Use Core ML for image generation where ANE/Core ML compile path is better.
- Keep one heavy model resident at a time: SLM, VLM, or diffusion.

6. C — MacBook/macOS proposal, with MLX optimization where possible

MacBook is the strongest Apple target for local generation. Use **MLX-LM** for text generation, **Core ML or ONNX Runtime CoreML EP** for classifiers/embeddings, and **Core ML / Diffusers MPS / MLX examples** for image generation depending on packaging. MLX-LM supports conversion, quantization, and streaming generation. ([GitHub](#))

C1. MacBook low tier

Target device

Parameter	Target
RAM	8 GB unified memory
Storage headroom	2–8 GB
Compute	M1/M2 Air-class or Intel legacy
Default mode	strong text generation; limited creative

Core models

Capability	Apple Silicon low	Intel Mac low
Text guardrail/search	XLM-R INT4/INT8 Core ML/ONNX	XLM-R INT4/INT8 ONNX CPU
Text generation	prism-ml/Ternary-Bonsai-1.7B-mlx-2bit	Qwen/Qwen3-0.6B GGUF or Bonsai GGUF if benchmark passes
Image safety	MobileNetV3-Small + TinyViT-5M Core ML	ONNX CPU
OCR	Apple Vision	Apple Vision
Speech-to-text	Whisper Tiny/Base	Whisper Tiny/Base CPU
Media search	MobileCLIP-S2/Core ML	MobileCLIP-S2/ONNX
Visual Q&A	SmolVLM2 optional only	disabled or experimental
TTS	Kokoro ONNX or system TTS	Kokoro ONNX or system TTS

Generation add-ons

Add-on	Apple Silicon low	Intel Mac low
Image generation	stabilityai/sd-turbo optional	disabled by default
Music/vocal generation	stable-audio-open-small optional only	disabled
Video generation	disabled	disabled

Optimization

- Apple Silicon: MLX for SLM.
- Intel: treat as CPU-only Windows class.
- Cap text generation to 2K–4K context.
- No default video or music pack.

C2. MacBook mid tier

Target device

Parameter	Target
RAM	16–24 GB unified memory
Storage headroom	10–50 GB
Compute	M-series GPU, stable MLX
Default mode	full KChat core + image add-on + limited audio add-on

Core models

Capability	Model	Format	Quantization Runtime	
Text guardrail/search	KChat XLM-R	Core ML/ONNX	INT8	Core ML / ONNX Runtime
Text generation	prism-ml/Ternary-Bonsai-1.7B-mlx-2bit	MLX	2-bit	MLX-LM
Optional quality SLM	Qwen/Qwen3-4B-Instruct-2507	MLX	4-bit	opt-in
Image safety	MobileNetV3-Small + TinyViT-5M	Core ML	INT8	Core ML
OCR	Apple Vision	system	n/a	Vision
Speech-to-text	openai/whisper-base	MLX/Core ML/whisper.cpp	INT8/FP16	benchmark-selected

Capability	Model	Format	Quantization Runtime	
Media search	apple/MobileCLIP2-S2	Core ML	INT8	lazy
Visual Q&A	SmolVLM2-500M-Video-Instruct	MLX/VLM candidate	4-bit/8-bit	optional
TTS	onnx-community/Kokoro-82M-v1.0-ONNX	ONNX	q8/q4	optional

Generation add-ons

Add-on	Model	Decision
Image generation	stabilityai/sd-turbo, stabilityai/sd-xl-turbo	allowed
Music/audio lite	stabilityai/stable-audio-open-small	allowed
Music + vocal	ACE-Step/ACE-Step-v1-3.5B	experimental
Full-song research	HKUSTAudio/YuE-s2-1B-general	disabled by default
Video generation	Lightricks/LTX-Video	experimental only; prefer API mode if local unsupported

Optimization

- MLX-LM for SLM.
- Core ML for image safety and embeddings.
- Use one heavy generator at a time.
- Creative models should run in a separate worker process.
- Provide progressive download: image first, audio second, video last.

C3. MacBook high-end tier

Target device

Parameter	Target
RAM	32–64 GB+ unified memory
Storage headroom	50–200 GB
Compute	M Pro/Max/Ultra-class GPU
Default mode	full local AI workstation except large video remains benchmark-gated

Core models

Capability	Model	Runtime
Text guardrail/search	XLM-R INT8	Core ML / ONNX Runtime
Text generation	prism-ml/Ternary-Bonsai-1.7B-mlx-2bit	MLX-LM
Higher-quality text	Qwen/Qwen3-4B-Instruct-2507	MLX 4-bit
Image safety	MobileNetV3-Small + TinyViT-5M	Core ML
OCR	Apple Vision	system
ASR	Whisper Base; optional Whisper Small later	MLX/Core ML/whisper.cpp
Media search	MobileCLIP2-S2	Core ML
VLM understanding	SmolVLM2 or Gemma 3n E2B	MLX/VLM candidate
TTS	Kokoro ONNX	ONNX

Generation add-ons

Add-on	Model	Decision
Image generation	stabilityai/sd-xl-turbo; optional SD3/SDXL Core ML conversions	allowed
Audio lite	stabilityai/stable-audio-open-small	allowed
Music + vocal	ACE-Step/ACE-Step-v1-3.5B	allowed after benchmark

Add-on	Model	Decision
Full-song research	HKUSTAudio/YuE-s2-1B-general, m-a-p/YuE-s1-7B-anneal-en-cot	research/pro only
Video generation	Lightricks/LTX-Video	experimental; local support depends on model/runtime
Heavy video	Wan2.2/Hunyuan	not recommended on MacBook unless separately validated

Optimization

- Use MLX streaming generation for chat.
- Use MLX 4-bit conversion for optional Qwen3-4B.
- Use Core ML Stable Diffusion conversion for image generation when targeting ANE/GPU.
- Use MPS/PyTorch only for experimental creative packs.
- Add “creative workspace” storage controls because video/music models can exceed tens of GB.

7. D — Windows laptop proposal: CPU-only vs CPU+GPU

Windows must not be treated as one class. It needs two separate execution paths:

1. Windows CPU-only

- ONNX Runtime CPU EP / XNNPACK / OpenVINO where applicable.
- llama.cpp CPU for GGUF text generation.
- No default image/music/video generation.

2. Windows CPU+GPU

- ONNX Runtime DirectML for classifiers/embeddings where stable.
- CUDA preferred for creative generation when NVIDIA GPU is available.
- DirectML can support some ONNX workloads, but most open creative-generation stacks are CUDA-first.

ONNX Runtime supports CPU, DirectML, Core ML, XNNPACK, CUDA, TensorRT, OpenVINO, and other execution providers, but CPU fallback must remain mandatory on Windows laptops. ([ONNX Runtime](#))

D1. Windows low tier

D1a. Windows low tier — CPU-only

Parameter	Target
RAM	4–8 GB
Storage headroom	1–3 GB
Compute	AVX2 preferred; no useful GPU
Default mode	text understanding, light text generation, speech optional

Capability	Model	Format	Quantization	Runtime
Text guardrail/search	KChat XLM-R	ONNX	INT4	ONNX Runtime CPU EP
Text generation	Qwen/Qwen3-0.6B	GGUF	Q4_K_M	llama.cpp CPU
Text generation alternative	prism-ml/Ternary-Bonsai-1.7B-gguf	GGUF	Q2_0	only if benchmark passes
Image safety	MobileNetV3-Small	ONNX	INT8	CPU EP
Enhanced safety	none	n/a	n/a	disabled
OCR	PaddlePaddle/PP-OCRv5_mobile_det + PP-OCRv5 mobile recognition pack, or existing KChat Windows OCR pack	ONNX	INT8	CPU EP

Capability	Model	Format	Quantization	Runtime
Speech-to-text	openai/whisper-tiny	ONNX / whisper.cpp	INT8	CPU
Media search	apple/MobileCLIP-S2	ONNX	INT4	lazy
TTS	onnx-community/Kokoro-82M-v1.0-ONNX	ONNX	q8/q4	optional
Image/music/video generation	none	n/a	n/a	disabled

PaddleOCR’s PP-OCRv5 mobile detector is suitable as a Windows OCR fallback because the mobile detector is designed for edge deployment and small model size; PaddleOCR also documents PP-OCRv5 mobile CPU deployment for on-device OCR.

(paddlepaddle.github.io)

D1b. Windows low tier — CPU+GPU

Parameter	Target
RAM	8 GB
GPU	iGPU or very low VRAM GPU

Default mode same as CPU-only; GPU used only for ONNX acceleration

Capability	Model	Runtime
Text generation	Qwen3-0.6B or Bonsai GGUF	llama.cpp CPU/Vulkan only if stable
Encoders/classifiers	XLM-R, MobileNetV3, MobileCLIP	ONNX Runtime DirectML or CPU fallback
Creative add-ons	none	disabled

Generation add-ons

Add-on	CPU-only	CPU+GPU
Image generation	disabled	disabled
Music/vocal generation	disabled	disabled

Add-on	CPU-only	CPU+GPU
Video generation	disabled	disabled

D2. Windows mid tier

D2a. Windows mid tier — CPU-only

Parameter	Target
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RAM 8–16 GB

Storage headroom 3–10 GB

Compute AVX2 or AVX-512/VNNI if available

Default mode full text core, no creative generation

Capability	Model	Format	Quantization Runtime	
Text guardrail/search	KChat XLM-R	ONNX	INT4/INT8	CPU EP
Text generation	prism-ml/Ternary-Bonsai-1.7B-gguf	GGUF	Q2_0	llama.cpp CPU
Fallback SLM	Qwen/Qwen3-0.6B	GGUF	Q4_K_M	llama.cpp CPU
Image safety	MobileNetV3-Small + TinyViT-5M	ONNX	INT8	CPU EP
OCR	KChat Windows OCR pack or PP-OCrv5 mobile OCR	ONNX	INT8	CPU EP
Speech-to-text	openai/whisper-base if benchmark passes; otherwise Tiny	ONNX / whisper.cpp	INT8	CPU
Media search	MobileCLIP-S2 or MobileCLIP2-S2	ONNX	INT8	CPU EP

Capability	Model	Format	Quantization	Runtime
Visual Q&A	SmolVLM2	ONNX/GGUF candidate	4-bit	optional, slow
TTS	Kokoro ONNX	ONNX	q8/q4	optional

D2b. Windows mid tier — CPU+GPU

Parameter	Target
RAM	16 GB
GPU	NVIDIA 4–8 GB VRAM, or DirectML-capable GPU
Storage headroom	10–50 GB
Default mode	full core + image generation; limited audio/video only on NVIDIA

Capability	Model	Runtime
Text generation	Bonsai GGUF	llama.cpp CPU/CUDA/Vulkan
Text/vision encoders	XLM-R, MobileCLIP2, TinyViT	ONNX Runtime DirectML/CUDA/CPU
Image generation	stabilityai/sd-turbo	CUDA preferred; DirectML only if validated
Music/audio lite	stabilityai/stable-audio-open-small	CUDA preferred
Video lite	Wan-AI/Wan2.1-T2V-1.3B or FramePack	NVIDIA only, benchmark-gated

Generation add-ons

Add-on	CPU-only	CPU+GPU
Image generation	disabled	sd-turbo, optional sdxl-turbo

Add-on	CPU-only	CPU+GPU
Music/vocal generation	disabled	Stable Audio Open Small; ACE-Step experimental
Video generation	disabled	Wan2.1-T2V-1.3B if ≥ 8 GB VRAM; FramePack if NVIDIA ≥ 6 GB VRAM

D3. Windows high-end tier

D3a. Windows high tier — CPU-only

Parameter	Target
RAM	16–32 GB
Storage headroom	10–30 GB
Compute	high-end CPU, AVX-512/VNNI preferred
Default mode	full text, speech, OCR, media search; no heavy creative

Capability	Model	Format	Quantization	Runtime
Text guardrail/search	XLM-R	ONNX	INT8	CPU EP / OpenVINO if validated
Text generation	Bonsai-1.7B	GGUF	Q2_0	llama.cpp CPU
Optional quality SLM	Qwen3-4B	GGUF	Q4	only if benchmark passes
Image safety	MobileNetV3 + TinyViT	ONNX	INT8	CPU EP
OCR	PP-OCrv5 mobile / KChat OCR pack	ONNX	INT8	CPU EP

Capability	Model	Format	Quantization Runtime	
ASR	Whisper Base	ONNX / whisper.cpp	INT8	CPU
Media search	MobileCLIP2-S2	ONNX	INT8	CPU EP
TTS	Kokoro ONNX	ONNX	q8/q4	CPU

D3b. Windows high tier — CPU+GPU

Parameter	Target
RAM	32–64 GB
GPU	NVIDIA RTX 8–24 GB+ VRAM
Storage headroom	50–200 GB
Default mode	full local KChat + all creative add-ons gated by VRAM

Capability	Model	Runtime
Text generation	Bonsai GGUF; optional Qwen3-4B	llama.cpp CUDA/Vulkan/CPU
Text understanding	XLM-R	ONNX Runtime CUDA/DirectML/CPU
Image safety/search	TinyViT, MobileCLIP2	ONNX Runtime CUDA/DirectML
ASR	Whisper Base	CUDA/CPU
TTS	Kokoro ONNX	ONNX Runtime
VLM understanding	SmolVLM2; candidate Gemma 3n	CUDA/ONNX/GGUF
Image generation	SD-Turbo, SDXL-Turbo	CUDA
Music/vocal	Stable Audio Open Small, ACE-Step-v1-3.5B, YuE research	CUDA

Capability	Model	Runtime
Video	Wan2.1, FramePack, HunyuanVideo, LTX-Video, Wan2.2	CUDA, VRAM-gated
Generation add-ons		
Add-on	CPU-only	CPU+GPU
Image generation	disabled by default	sd-turbo, sdxl-turbo
Music/vocal generation	disabled by default	stable-audio-open-small, ACE-Step/ACE-Step-v1-3.5B, optional YuE
Video generation	disabled	Wan-AI/Wan2.1-T2V-1.3B, llyasviel/FramePack, hunyuanvideo-community/HunyuanVideo, Lightricks/LTX-Video
Workstation video	disabled	Wan-AI/Wan2.2-TI2V-5B, Wan-AI/Wan2.2-T2V-A14B only if VRAM is sufficient

HunyuanVideo is a 13B-parameter diffusion transformer and should be restricted to high-end GPU devices. LTX-Video provides 2B and 13B variants and FP8/distilled variants, but local desktop deployment still depends heavily on VRAM and CUDA support. ([Hugging Face](#))

8. Add-on strategy

8.1 Image generation add-on

Tier	Android	iOS	MacBook	Windows CPU-only	Windows CPU+GPU
Low	disabled	disabled	optional only on Apple Silicon	disabled	disabled
Mid	sd-turbo experimental	sd-turbo Core ML optional	sd-turbo, sdxl-turbo	disabled	sd-turbo

Tier	Android	iOS	MacBook	Windows CPU-only	Windows CPU+GPU
High	sd-turbo; sdxl-turbo experimental	sd-turbo, sdxl-turbo	sdxl-turbo	disabled	sdxl-turbo

Actual models

- stabilityai/sd-turbo
- stabilityai/sdxl-turbo
- Apple deployment: Core ML Stable Diffusion conversion and Swift inference.

Optimization

- Use 512×512 default.
- Use one-to-four-step generation.
- Use FP16 first.
- Use Core ML palettization/activation quantization only after PSNR and visual regression tests.
- Reuse KChat media safety classifier instead of shipping a duplicate safety checker.
- Do not load image generation while SLM/VLM is resident on mobile.

8.2 Music + vocal generation add-on

Tier	Android	iOS	MacBook	Windows CPU-only	Windows CPU+GPU
Low	disabled	disabled	disabled except TTS	TTS only	TTS only
Mid	disabled	disabled	Stable Audio Open Small; ACE-Step experimental	TTS only	Stable Audio Open Small; ACE-Step experimental
High	disabled	disabled by default	ACE-Step; YuE research	TTS only	ACE-Step; YuE research

Actual models

Function	Model
Short audio / loop	stabilityai/stable-audio-open-small
Full music + vocals	ACE-Step/ACE-Step-v1-3.5B
Full-song research	HKUSTAudio/YuE-s2-1B-general
Full-song stage-1 research	m-a-p/YuE-s1-7B-anneal-en-cot
TTS / narration	onnx-community/Kokoro-82M-v1.0-ONNX or hexgrad/Kokoro-82M

Product rules

- TTS is not vocal singing.
- Vocal generation is desktop-first.
- Reference voice cloning must be disabled until consent and rights handling exist.
- Lyrics prompts pass through text guardrail.
- Audio output metadata should mark generated content.
- Export should be explicit.

8.3 Video generation add-on

Tier	Android iOS	MacBook	Windows CPU-only	Windows CPU+GPU
Low	disabled disabled	disabled	disabled	disabled
Mid	disabled disabled	experimental only	disabled	Wan2.1 / FramePack if GPU permits
High	disabled disabled by default	experimental LTX only	disabled	Wan2.1, FramePack, HunyuanVideo, LTX
Workstation	n/a n/a	not primary	n/a	Wan2.2, LTX 13B, Hunyuan

Actual models

Function	Model
Text-to-video lite	Wan-AI/Wan2.1-T2V-1.3B
Image-to-video low-VRAM desktop	llyasviel/FramePack
High-end video	hunyuanvideo-community/HunyuanVideo
Image-to-video pro	Lightricks/LTX-Video
Workstation video	Wan-AI/Wan2.2-TI2V-5B, Wan-AI/Wan2.2-T2V-A14B

Product rules

- No video generation on Windows CPU-only.
- No default video generation on mobile.
- NVIDIA CUDA path is the first production target.
- DirectML video generation is research only unless model-specific validation proves stability.
- Store models outside the core KChat model directory.
- Require disk quota, resumable downloads, and user-visible model size warnings.

9. Proposed delivery phases

Phase 0 — embedded install

Artifact	Platforms	Size target
MobileNetV3-Small INT8	all	<8 MB
KChat skill pack	all	<1 MB
rule-based guardrail fallback	all	<1 MB

Phase 1 — first launch core

Artifact	Low	Mid	High
XLM-R	INT4	INT4/INT8	INT8
Text generator	Qwen3-0.6B or Bonsai if benchmark passes	Bonsai-1.7B	Bonsai-1.7B
TinyViT-5M	no	yes	yes
OCR pack	Windows only if needed	Windows only if needed	Windows only if needed

Phase 2 — on-demand understanding

Trigger	Artifact
First voice input	Whisper Tiny/Base
First media search	MobileCLIP-S2 or MobileCLIP2-S2
First image/video Q&A	SmolVLM2-500M
First TTS use	Kokoro ONNX or system TTS
First OCR on Windows	PP-OCrv5 mobile OCR or existing KChat OCR pack

Phase 3 — creative add-ons

Trigger	Artifact
Enable image generation	SD-Turbo / SDXL-Turbo
Enable music generation	Stable Audio Open Small / ACE-Step
Enable full-song research	YuE
Enable video generation	Wan2.1 / FramePack / Hunyuan / LTX / Wan2.2

10. Benchmark gates

10.1 Required benchmark before model download

Benchmark	Gate
Available RAM check	decides SLM/VLM/diffusion eligibility
Free storage check	prevents creative add-on download failure
128-token prefill + 32-token decode	selects Qwen3-0.6B vs Bonsai
XLNet batch inference	selects INT4 vs INT8
3-image scan	enables/disables TinyViT
10-second audio transcription	selects Whisper Tiny vs Base
512×512 image dry run	enables image generation
GPU/VRAM probe	enables music/video generation
thermal/battery check	mobile background generation control

10.2 Minimum acceptable thresholds

Feature	Minimum target
Smart reply	<3 seconds perceived latency
Summary	streaming starts <5 seconds
Text guardrail	<250 ms p95 for short text
Image safety	<100 ms low/mid; <50 ms high
Whisper Tiny	faster than real time for short clips
SD-Turbo image	acceptable only if 512×512 completes without app termination
Music generation	desktop only; must show progress
Video generation	desktop GPU only; must show progressive preview

11. Memory and storage budgets

11.1 Core storage target

Platform/tier	Core target
Android low	150–700 MB
Android mid	500 MB–1.5 GB
Android high	1–2 GB
iOS low	150–700 MB
iOS mid	500 MB–1.5 GB
iOS high	1–2 GB
MacBook low	1–3 GB
MacBook mid	2–6 GB
MacBook high	3–10 GB
Windows low CPU-only	1–3 GB
Windows mid CPU-only	2–6 GB
Windows high CPU-only	3–10 GB
Windows GPU mid/high	3–10 GB core

11.2 Add-on storage target

Add-on	Typical target
SD-Turbo	1–3 GB
SDXL-Turbo	4–8 GB
Stable Audio Open Small	2–6 GB
ACE-Step	5–20 GB
YuE	10–50 GB
Wan2.1-T2V-1.3B	10–20 GB
FramePack	>30 GB

Add-on	Typical target
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HunyuanVideo	20–60 GB
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LTX / Wan2.2	50–200 GB
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FramePack’s own README states that its Windows package downloads more than 30 GB from Hugging Face, so KChat should never bundle it into the default install. ([GitHub](#))

12. Final recommended matrix

A. Android

Tier	Core generation	Understanding	Add-ons
Low	Qwen3-0.6B Q4; Bonsai only if benchmark passes	XML-R INT4, MobileNetV3, ML Kit OCR, Whisper Tiny	none
Mid	Bonsai-1.7B GGUF	XML-R, TinyViT, Whisper Tiny/Base, MobileCLIP, optional SmolVLM2	SD-Turbo experimental
High	Bonsai-1.7B; optional Qwen3-4B	full core + SmolVLM2/Gemma 3n candidate	SD-Turbo; SDXL-Turbo experimental

B. iOS

Tier	Core generation	Understanding	Add-ons
Low	Qwen3-0.6B; Bonsai MLX only if benchmark passes	XML-R, MobileNetV3, Apple Vision, Whisper Tiny	none
Mid	Bonsai-1.7B MLX	XML-R, TinyViT, Apple Vision, Whisper, MobileCLIP	SD-Turbo Core ML optional
High	Bonsai-1.7B MLX; optional Qwen3-4B MLX	full core + SmolVLM2/Gemma 3n candidate	SD-Turbo / SDXL-Turbo Core ML

C. MacBook

Tier	Core generation	Understanding	Add-ons
Low	Bonsai MLX on Apple Silicon; Qwen3-0.6B on Intel	XLM-R, TinyViT, Apple Vision, Whisper, MobileCLIP	SD-Turbo only on Apple Silicon
Mid	Bonsai MLX; optional Qwen3-4B	full core + SmolVLM2	SDXL-Turbo, Stable Audio, ACE-Step experimental
High	Bonsai MLX + Qwen3-4B optional	full core + VLM candidate	image, audio, ACE-Step, YuE research, LTX experimental

D. Windows laptop

Tier	CPU-only	CPU+GPU
Low	Qwen3-0.6B; XLM-R INT4; MobileNetV3; Whisper Tiny; no creative	same core; DirectML for ONNX if useful; no creative
Mid	Bonsai if benchmark passes; no creative except TTS	Bonsai; SD-Turbo; Stable Audio; Wan2.1/FramePack only if NVIDIA VRAM permits
High	Bonsai; optional Qwen3-4B if CPU is strong; no heavy creative	full core + SDXL-Turbo + ACE-Step + Wan2.1/FramePack/Hunyuan/LTX by VRAM

13. Final architecture decision

KChat should ship a **minimal mandatory core** and make all heavy generation optional.

Mandatory core

Text understanding:

KChat XLM-R INT4/INT8 ONNX

xlmr.spm

xlmr_head.npz

Text generation:

Android/Windows: prism-ml/Ternary-Bonsai-1.7B-gguf

iOS/Mac Apple Silicon: prism-ml/Ternary-Bonsai-1.7B-mlx-2bit

low fallback: Qwen/Qwen3-0.6B

Media safety:

MobileNetV3-Small INT8

timm/tiny_vit_5m_224.dist_in22k_ft_in1k on mid/high

Speech:

openai/whisper-tiny

openai/whisper-base

Media search:

apple/MobileCLIP-S2

candidate upgrade: apple/MobileCLIP2-S2

Optional media understanding:

HuggingFaceTB/SmolVLM2-500M-Video-Instruct

candidate replacement: google/gemma-3n-E2B-it

Optional TTS:

onnx-community/Kokoro-82M-v1.0-ONNX

Optional creative packs

Image:

stabilityai/sd-turbo

stabilityai/sd-xl-turbo

Music + vocal:

stabilityai/stable-audio-open-small

ACE-Step/ACE-Step-v1-3.5B

HKUSTAudio/YuE-s2-1B-general

m-a-p/YuE-s1-7B-anneal-en-cot

Video:

Wan-AI/Wan2.1-T2V-1.3B

llyasviel/FramePack

hunyuanvideo-community/HunyuanVideo

Lightricks/LTX-Video

Wan-AI/Wan2.2-TI2V-5B

Wan-AI/Wan2.2-T2V-A14B

Final rule

The product should not install more than one default text generator, one media embedding model, or one VLM at the same time. Creative add-ons must be isolated, optional, and benchmark-gated. This keeps KChat small for normal users while still enabling advanced local generation on hardware that can actually run it.